MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 0B2WA

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Voltage regulator, miniature

Outline	---	5-3 (EIA)
Base	---	E7-1
Envelope	---	T5-1/2
Cathode	---	Glow discharge

Base connections:

Pin No.	---	1	2	3	4	5	6	7
Element	---	a	k	int con	k	a	int con	k

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Total darkness ionization voltage	Ambient light ionization voltage	Operating voltage	Operating current	TA	TE	Altitude
Unit:	Vdc	Vdc	Vdc	mAdc	°C	°C	ft
Maximum:	---	---	116	30	---	150	See note 1
Minimum:	130	130	103	5	-55	---	---
TEST CONDITIONS:	---	---	---	---	---	---	---

GENERAL:

Qualification - Required

Reliable tubes

Ⓖ denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
3347	Ionization voltage (1) (with illumination)	Ebb/Ib = 5 to 30 mAdc; illumination = 5 to 50 ft candles (see note 2)	0.4	II	Ez	---	130	Vdc
3337	Voltage drop (1)	Ebb/Ib = 30 mAdc	0.4	II	Etd	105	111	Vdc
3337	Voltage drop (2)	Ebb/Ib = 5 mAdc	0.4	II	Etd	105	111	Vdc
3335	Regulation	(1) Etd - (2) Etd	0.4	II	---	---	±2.5	Vdc
1201	Short and discontinuity detection		0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
1031	Low-frequency vibration	Rp = 10,000 ohms; Ebb/Ib = 20 mAdc	2.5	I	Ep	---	100	mVac
(G) 3347	Ionization voltage (2)	See note 3	2.5	I	Ez	---	130	Vdc
3337	Voltage drop (3)	Ebb/Ib = 20 mAdc	2.5	I	Etd	105	111	Vdc
3305	Leakage current	Eb = 50 Vdc; Rp = 3,000 ohms	2.5	I	LIb	---	5	µAdc
3278	Noise and oscillation (oscillation)	Esig = 100 mVac; Ebb/Ib = 5 to 30 mAdc	1.0	I	---	---	---	---
3278	Noise and oscillation (noise)	Ebb/Ib = 30 mAdc	1.0	I	ΔEB	---	5	mVac
---	Voltage jump	Ebb/Ib = 5 to 30 mAdc (see note 4)	2.5	I	---	---	100	mVdc
---	Repeatability	Ebb/Ib = 10 mAdc (see note 5)	2.5	I	---	---	600	mVdc
1121	Base strain		---	---	---	---	---	---
(G) 1126	Glass strain		2.5	I	---	---	---	---
(G) 1041	Shock test	900 G	15	S4	---	---	---	---
1031	Vibration fatigue	2.5 G; fixed frequency; F = 25 min, 60 max	2.5	See note 6	---	---	---	---
---	Shock and vibration-fatigue test end points:							
1031	Low-frequency vibration		---	---	Ep	---	100	mVac
3347	Ionization voltage (1)		---	---	Ez	---	130	Vdc
3337	Voltage drop (1)		---	---	Etd	103	113	Vdc
3337	Voltage drop (2)		---	---	Etd	103	113	Vdc
3335	Regulation		---	---	---	---	±3	Vdc
1105	Permanence of marking		---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP LEVEL OR CODE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC		SYMBOL	LIMITS		UNIT
					1ST SAMPLE	COMBINED SAMPLE		MIN	MAX	
	Quality conformance inspection, part 3									
Ⓖ1516	Stability life	Ebb/Ib = 20 mAdc; TA = room	---	---	---	---	---	---	---	---
---	Stability life-test end point (2 hours):									
---	Change in voltage drop (3) of individual tubes		---	---	---	---	ΔEtd t	---	2.0	Vdc
1521	Survival-rate life	Stability life test, or equivalent conditions	---	---	---	---	---	---	---	---
---	Survival-rate life-test end points:									
1201	Short and discontinuity detection		---	---	---	---	---	---	---	---
---	Change in voltage drop (3) of individual tubes		---	---	---	---	ΔEtd t	---	3.0	Vdc
Ⓖ1501	Life test	Stability life test, or equivalent conditions: TE = 150°C (min) (see note 7)	---	---	---	---	---	---	---	---
---	Life-test end points (500 hours):									
---	Inoperatives		---	---	1	3	---	---	---	---
3335	Regulation		---	---	1	3	---	---	±3	Vdc
3337	Voltage drop (1)		---	---	1	3	Etd	103	113	Vdc
3337	Voltage drop (2)		---	---	1	3	Etd	103	113	Vdc
3337	Voltage drop (3)		---	---	1	3	Etd	103	113	Vdc
---	Change in voltage drop (3) of individual tubes		---	---	1	3	ΔEtd t	---	4.0	Vdc
3347	Ionization voltage (1)		---	---	1	3	Ez	---	130	Vdc
---	Total defectives		---	---	4	8	---	---	---	---
---	Life-test end points (1,000 hours):									
---	Inoperatives		---	---	2	5	---	---	---	---
3335	Regulation		---	---	2	5	---	---	±4	Vdc
3337	Voltage drop (1)		---	---	2	5	Etd	103	116	Vdc
3337	Voltage drop (2)		---	---	2	5	Etd	103	116	Vdc
3337	Voltage drop (3)		---	---	2	5	Etd	103	116	Vdc
---	Change in voltage drop (3) of individual tubes		---	---	2	5	ΔEtd t	---	5.0	Vdc
3347	Ionization voltage (1)		---	---	2	5	Ez	---	130	Vdc
---	Total defectives		---	---	5	10	---	---	---	---

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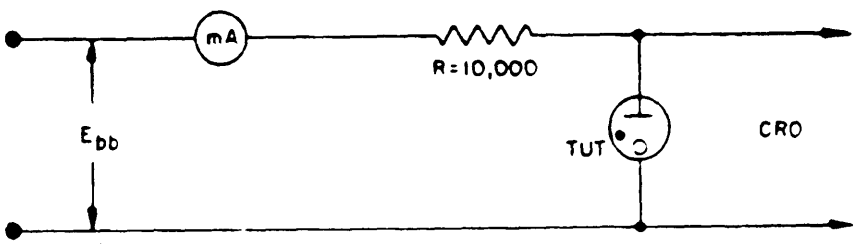
NOTES:

1. See "Reduced pressure (altitude) rating" and altitude, maximum peak voltage in the basic document.

2. This test to be performed at the conclusion of the holding period.

Ⓖ 3. The tube shall fire within 20 seconds maximum.

4. Vary current by using the following circuit, from 5 to 30 mAdc and back to 5 mAdc (by adjusting Ebb slowly). Sudden voltage jumps registered on the oscilloscope shall be not greater than the specified value.



5. The tube shall be tested in the following manner:

   (a) The voltage drop shall be read at 10 mAdc drain.
   (b) The tube shall be turned off for 1 minute.
   (c) The tube shall be restarted and operated at the same current.
   (d) Etd shall be read after 1 minute of operation.
   (e) The on-off cycle shall be repeated a minimum of five times. The maximum difference in voltage drop shall be taken as the measure of repeatability.

6. This test shall be conducted on the initial lot and thereafter on a lot approximately every 6 months. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. When one lot passes, the 6-month rule shall apply. MIL-STD-105, sample code letter E, shall apply.

Ⓖ 7. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (see method 1226), will be satisfied if a tube having bogey Ib (±3 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.

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Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MI, MU
Air Force - 80
DSA - ES

User activities:
Army - ME
Navy - AS, OS, MC, CG
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-3049)

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